NAZON

FLEET TRACKING ANALYSER

Python Mastery Guide

Beginner → Intermediate → Pro

Roads · Great East · Cairo · Kafue · Great North · Mumbwa

Coverage · Lusaka · Copperbelt · All Zambia · East Africa

Mode · Real-Time Tracking — Day & Night Intelligence

AI Engine · Claude AI by Anthropic

Platform · GPS + Google Maps API + OpenStreetMap

Lessons · 20 Full Lessons + Dozens of Runnable Examples

Every code block is ready to run in VS Code. Build a live fleet safety system from scratch.

"The future depends on what you do today."

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BEGINNER

Python Foundations for NAZON Fleet Work through every example. Type them yourself. That's how it sticks.

LESSON 01

Variables & Data Types

NAZON Fleet uses variables for vehicle IDs, GPS coordinates, speeds and road alerts.

A variable is a labelled box. You give it a name and store a value inside it. Fleet tracking needs dozens of variables for every vehicle on the road.

Example 1 — Your First Fleet Variables

```
# Create variables

system_name  = 'NAZON Fleet'

version      = 1

total_vehicles = 12

tracking_live  = True


# Print them

print(system_name)

print(version)

print(total_vehicles)

print(tracking_live)
```

```
NAZON Fleet

1

12

True
```

Example 2 — Vehicle Tracking Variables

```
# Real variables NAZON Fleet uses for every vehicle

vehicle_id   = 'NZ-001'          # unique vehicle code

driver_name  = 'Mwila Banda'     # driver on duty

plate_number = 'ABZ 1234 ZM'     # licence plate

latitude     = -15.4167          # GPS latitude (Lusaka)

longitude    = 28.2833           # GPS longitude (Lusaka)

speed_kmh    = 65.5              # current speed in km/h

road_name    = 'Great East Road' # current road

is_night     = False             # is it night time?

alert_level  = 'GREEN'           # GREEN / AMBER / RED
```

```
print(vehicle_id)
print(driver_name)
print(road_name)
print(speed_kmh)
```

```
NZ-001
Mwila Banda
Great East Road
65.5
```

Example 3 — Updating Variables

```
# Speed changes as the vehicle moves
speed_kmh = 65.5
print(speed_kmh)

# Driver hits traffic — slows down
speed_kmh = speed_kmh - 20.0
print(speed_kmh)

# Shorthand
speed_kmh -= 5.0
print(speed_kmh)

# Night begins
is_night = False
is_night = True
print(is_night)
```

```
65.5
45.5
40.5
True
```

Example 4 — The 4 Data Types

```
# str — text (in quotes)
vehicle_id  = 'NZ-001'
road_name   = 'Kafue Road'
```

```
alert_level = 'RED'


# float — decimal number

latitude    = -15.4167

longitude   = 28.2833

speed_kmh   = 65.5


# int — whole number

vehicles_on_road = 12

incidents_today  = 3

congestion_index = 7   # 1-10 scale


# bool — True or False only

tracking_live   = True

is_night        = False

danger_road     = True


print(type(vehicle_id))

print(type(latitude))

print(type(vehicles_on_road))

print(type(tracking_live))
```

```
<class 'str'>

<class 'float'>

<class 'int'>

<class 'bool'>
```

Example 5 — Math with Variables (Speed & Distance)

```
distance_km  = 45.0   # km from Lusaka to destination

speed_kmh    = 80.0   # average speed

fuel_level   = 0.65   # 65% full

tank_km      = 500.0  # full tank range in km


eta_hours  = distance_km / speed_kmh

eta_mins   = eta_hours * 60

fuel_range = fuel_level * tank_km
```

```
print(f'ETA       : {round(eta_mins, 1)} minutes')
print(f'Fuel left : {round(fuel_range)} km range')
print(f'Enough fuel: {fuel_range > distance_km}')
```

```
ETA       : 33.8 minutes
Fuel left : 325 km range
Enough fuel: True
```

✏ PRACTICE

1. Create variables: vehicle_id, driver_name, plate_number, road_name for one vehicle.
2. Create GPS coords for Lusaka: latitude=-15.4167, longitude=28.2833. Print them.
3. Create speed_kmh=90.0. Subtract 25 using -=. Print before and after.
4. Create 4 data types: str (road), float (lat), int (vehicles), bool (is_night). Print each with type().
5. Calculate ETA: distance=120.0 km, speed=90.0 km/h. Print ETA in minutes.

LESSON 02

Strings & F-Strings

NAZON Fleet uses strings for driver alerts, AI messages, road warnings and SMS notifications.

Example 1 — String Basics

```
road1 = 'Great East Road'

road2 = "Kafue Road"

alert = 'Heavy traffic ahead!'

print(road1)

print(road2)

print(alert)
```

```
Great East Road

Kafue Road

Heavy traffic ahead!
```

Example 2 — Joining Strings (Use F-Strings!)

```
driver = 'Mwila Banda'

road   = 'Cairo Road'

status = 'CONGESTED'


# Old way — messy

msg = 'Alert for ' + driver + ': ' + road + ' is ' + status

print(msg)


# Better way — f-strings (use this in NAZON Fleet!)

msg = f'Alert for {driver}: {road} is {status}'

print(msg)
```

```
Alert for Mwila Banda: Cairo Road is CONGESTED

Alert for Mwila Banda: Cairo Road is CONGESTED
```

Example 3 — F-Strings with Numbers

```
vehicle_id  = 'NZ-003'

speed_kmh   = 112.7

latitude    = -15.4167

longitude   = 28.2833

congestion  = 8
```

```
print(f'Vehicle   : {vehicle_id}')
print(f'Speed     : {speed_kmh:.1f} km/h')
print(f'Location  : ({latitude:.4f}, {longitude:.4f})')
print(f'Congestion: {congestion}/10')

# Speed alert formatting
over_limit = speed_kmh - 100
print(f'Over limit by: {over_limit:.1f} km/h')
```

```
Vehicle   : NZ-003
Speed     : 112.7 km/h
Location  : (-15.4167, 28.2833)
Congestion: 8/10
Over limit by: 12.7 km/h
```

Example 4 — Building the AI Road Intelligence Prompt (Real NAZON Code)

```
vehicle_id = 'NZ-001'
road       = 'Great East Road'
destination= 'Kenneth Kaunda International Airport'
speed_kmh  = 65.0
congestion = 7
is_night   = True
time_str   = "21:45"
weather    = 'Light rain'
distance_km= 18.5

prompt = f'''
You are NAZON Fleet AI, a road safety intelligence system for Zambia.
Analyse the following journey and advise the driver.

Vehicle    : {vehicle_id}
Road       : {road}
Destination: {destination}
Speed      : {speed_kmh} km/h
```

```
Congestion : {congestion}/10
Night Drive: {is_night}
Time       : {time_str}
Weather    : {weather}
Distance   : {distance_km} km

Provide: ROAD_STATUS, SAFETY_LEVEL, ADVICE (one sentence).
'''
print(prompt)
```

```
You are NAZON Fleet AI, a road safety intelligence system for Zambia.
Analyse the following journey and advise the driver.

Vehicle    : NZ-001
Road       : Great East Road
Destination: Kenneth Kaunda International Airport
Speed      : 65.0 km/h
Congestion : 7/10
Night Drive: True
Time       : 21:45
Weather    : Light rain
Distance   : 18.5 km

Provide: ROAD_STATUS, SAFETY_LEVEL, ADVICE (one sentence).
```

Example 5 — String Methods

```
road = ' great east road '
print(road.strip())         # remove spaces
print(road.strip().title()) # Title Case
print('kafue road'.upper()) # ALL CAPS

# Check road type
road_name = 'Great North Road'
print('North' in road_name)    # True
print('South' in road_name)    # False
```

```
# Parse alert from AI response

ai_reply = 'ROAD_STATUS:CONGESTED:SAFETY:AMBER'

parts = ai_reply.split(':')

print(parts)


# Driver phone number cleanup

phone = '+260 97 123 4567'

clean = phone.replace(' ', '')

print(clean)
```

```
great east road

Great East Road

KAFUE ROAD

True

False

['ROAD_STATUS', 'CONGESTED', 'SAFETY', 'AMBER']

+26097123456■7
```

✏ PRACTICE

1. Create road='great east road'. Print it in Title Case and UPPERCASE.
2. Use an f-string: 'Vehicle NZ-001 on Great East Road: speed 72.5 km/h, ETA 25 min'.
3. Build a 6-line AI prompt for a night drive on Kafue Road using f-strings.
4. Use .split(':') on 'ALERT:SPEED:DANGER:RED' and print each part.
5. Check if 'East' is in 'Great East Road' and 'North' is in 'Kafue Road' using 'in'.

LESSON 03

Lists

NAZON Fleet stores vehicle IDs, dangerous roads, route waypoints and speed readings in lists.

Example 1 — Creating and Accessing Lists

```
# All vehicles in the NAZON fleet
fleet = ['NZ-001', 'NZ-002', 'NZ-003', 'NZ-004', 'NZ-005']

# Access by index (starts at 0)
print(fleet[0])   # first
print(fleet[2])   # third
print(fleet[-1])  # last
print(len(fleet)) # total count
```

```
NZ-001
NZ-003
NZ-005
5
```

Example 2 — Dangerous Roads & Night Hotspots

```
# Roads flagged as dangerous at night in Lusaka & Zambia
danger_roads_night = [
    'Kafue Road',         # poorly lit, potholes
    'Great North Road',   # truck traffic, night robbery hotspot
    'Mumbwa Road',        # poor road surface past Katuba
    'Lusaka-Livingstone', # wildlife crossings at night
    'Ndola-Kitwe Rd',     # high accident rate
]

# Check if a road is dangerous
road = 'Kafue Road'
print(road in danger_roads_night)

# Add a newly flagged road
danger_roads_night.append('Los Angeles Blvd')
print(len(danger_roads_night))
```

```
# Remove when repaired
danger_roads_night.remove('Los Angeles Blvd')
print(len(danger_roads_night))
```

```
True
6
5
```

Example 3 — Speed Readings from a Vehicle

```
# Last 6 speed readings for NZ-001 (km/h)
speeds = [78.0, 82.5, 95.0, 110.2, 88.0, 72.5]

print('Max speed    :', max(speeds))
print('Min speed    :', min(speeds))
print('Latest speed :', speeds[-1])
avg = sum(speeds) / len(speeds)
print(f'Average speed: {round(avg,1)} km/h')

# Did the vehicle ever exceed 100 km/h?
over_100 = [s for s in speeds if s > 100]
print(f'Speeding events: {len(over_100)}')
```

```
Max speed    : 110.2
Min speed    : 72.5
Latest speed : 72.5
Average speed: 87.7 km/h
Speeding events: 1
```

Example 4 — Route Waypoints

```
# Planned waypoints for NZ-002 Lusaka → Chipata
route = [
    'Lusaka CBD',
    'Great East Road Toll',
    'Chongwe',
    'Rufunsa',
    'Petauke',
    'Katete',
```

```
    'Chipata',
]

print(f'Total stops: {len(route)}')
print(f'Start      : {route[0]}')
print(f'End        : {route[-1]}')
print(f'Next stop  : {route[2]}')

# Slice — first 3 waypoints only
print(route[:3])
```

```
Total stops: 7
Start      : Lusaka CBD
End        : Chipata
Next stop  : Chongwe
['Lusaka CBD', 'Great East Road Toll', 'Chongwe']
```

✏ PRACTICE

1. Create a fleet list with 5 vehicle IDs. Print index 0, -1, and total count.
2. Create danger_roads_night with 4 Zambian roads. Check if 'Kafue Road' is in it.
3. Create speeds=[70,85,102,95,88]. Find max, min, average. Count readings above 100.
4. Create a route list Lusaka → Livingstone with 5 stops. Slice first 3.
5. Add 'NZ-006' to fleet, print, then remove it and print again.

LESSON 04

Dictionaries

Every vehicle profile, road status, GPS ping and AI alert in NAZON Fleet is a dictionary.

Example 1 — Vehicle Profile Dictionary

```
vehicle = {
    'id'         : 'NZ-001',
    'plate'      : 'ABZ 1234 ZM',
    'driver'     : 'Mwila Banda',
    'phone'      : '+260971234567',
    'make'       : 'Toyota Hilux',
    'fuel_pct'   : 0.72,
    'speed_kmh'  : 65.5,
    'on_duty'    : True,
}

print(vehicle['driver'])
print(vehicle['plate'])
print(f"Speed: {vehicle['speed_kmh']} km/h")
```

```
Mwila Banda
ABZ 1234 ZM
Speed: 65.5 km/h
```

Example 2 — GPS Location Dictionary

```
# What NAZON receives from GPS every 30 seconds
gps_ping = {
    'vehicle_id' : 'NZ-001',
    'timestamp'  : '2026-05-12 21:45:30',
    'latitude'   : -15.4167,
    'longitude'  : 28.2833,
    'speed_kmh'  : 65.5,
    'heading_deg': 045,       # compass bearing
    'road'       : 'Great East Road',
    'accuracy_m' : 4.2,       # GPS accuracy metres
```

```
}

print(gps_ping['road'])
print(gps_ping['speed_kmh'])

# Update speed on next ping
gps_ping['speed_kmh'] = 72.0
print(gps_ping['speed_kmh'])
```

```
Great East Road
65.5
72.0
```

Example 3 — Road Status Dictionary (Core NAZON Data)

```
# road_analyser.py returns this for every major road
road_status = {
    'road'          : 'Great East Road',
    'section'       : 'Woodlands to Airport',
    'congestion'    : 8,         # 1-10 scale
    'avg_speed_kmh' : 35.0,      # actual traffic speed
    'speed_limit'   : 80,        # legal limit
    'is_night'      : True,
    'danger_level'  : 'AMBER',   # GREEN/AMBER/RED
    'hazards'       : ['potholes', 'poor lighting'],
    'ai_advice'     : 'Reduce speed, headlights on full beam.',
}

print(f"{road_status['road']} — {road_status['danger_level']}")
print(f"Congestion : {road_status['congestion']}/10")
print(f"Avg speed  : {road_status['avg_speed_kmh']} km/h")
print(f"Hazards    : {road_status['hazards']}")
print(road_status['ai_advice'])
```

```
Great East Road — AMBER
Congestion : 8/10
Avg speed  : 35.0 km/h
```

```
Hazards    : ['potholes', 'poor lighting']

Reduce speed, headlights on full beam.
```

Example 4 — Fleet-Wide Road Status (Nested Dictionaries)

```
# NAZON stores all road statuses in one dictionary

all_roads = {

    'Great East Road': {'congestion':8,'danger':'AMBER','speed_kmh':35},

    'Cairo Road'     : {'congestion':9,'danger':'RED',  'speed_kmh':15},

    'Kafue Road'     : {'congestion':4,'danger':'RED',  'speed_kmh':60},

    'Mumbwa Road'    : {'congestion':2,'danger':'AMBER','speed_kmh':75},

}


# Access nested value

print(all_roads['Cairo Road']['danger'])

print(all_roads['Kafue Road']['congestion'])


# Loop through all roads

for road, data in all_roads.items():

    print(f"{road}: {data['danger']} (congestion {data['congestion']}/10)")
```

```
RED

4

Great East Road: AMBER (congestion 8/10)

Cairo Road: RED (congestion 9/10)

Kafue Road: RED (congestion 4/10)

Mumbwa Road: AMBER (congestion 2/10)
```

✏ PRACTICE

1. Create a vehicle dict with id, driver, plate, speed_kmh, fuel_pct, on_duty.
2. Create a gps_ping dict with lat, lon, speed, road. Print driver and road.
3. Create road_status for Kafue Road with congestion, danger_level, hazards, ai_advice.
4. Print road, danger_level, and ai_advice using f-strings.
5. Create all_roads with 4 Zambian roads (nested). Loop and print each road's danger level.

LESSON 05

Conditionals (if / elif / else)

NAZON Fleet makes decisions. Is this road dangerous? Is the driver speeding? Should we send an alert?

Example 1 — Basic Speed Check

```
speed_kmh   = 112.5
speed_limit = 100

if speed_kmh > speed_limit + 20:
    print('CRITICAL: Dangerously over speed limit!')
elif speed_kmh > speed_limit:
    print('WARNING: Over speed limit!')
else:
    print('Speed OK.')
```

```
CRITICAL: Dangerously over speed limit!
```

Example 2 — Night Road Safety Logic (Real NAZON Code)

```
# road_analyser.py — checks every vehicle at night
road       = 'Kafue Road'
is_night   = True
speed_kmh  = 90.0
congestion = 3

danger_roads = [
    'Kafue Road', 'Great North Road',
    'Mumbwa Road', 'Lusaka-Livingstone',
]

if is_night and road in danger_roads:
    alert = 'RED'
    advice = 'DANGEROUS night road — reduce speed, stay alert.'
elif is_night:
    alert = 'AMBER'
    advice = 'Night driving — use full headlights.'
elif congestion >= 8:
```

```
    alert = 'AMBER'
    advice = 'Heavy traffic — allow extra journey time.'
else:
    alert = 'GREEN'
    advice = 'Road conditions normal. Safe travels.'

print(f'Alert : {alert}')
print(f'Advice: {advice}')
```

```
Alert : RED
Advice: DANGEROUS night road — reduce speed, stay alert.
```

Example 3 — Driver Unfamiliar with Lusaka Roads

```
driver_city     = 'Ndola'       # driver is from Copperbelt
current_city    = 'Lusaka'
familiar_cities = [
    'Ndola', 'Kitwe', 'Mufulira'
]

if current_city not in familiar_cities:
    print(f'NOTE: {driver_city} driver on unfamiliar roads.')
    print('Enabling enhanced navigation guidance.')
    enhanced_nav = True
else:
    print('Driver knows this area.')
    enhanced_nav = False

print(f'Enhanced nav: {enhanced_nav}')
```

```
NOTE: Ndola driver on unfamiliar roads.
Enabling enhanced navigation guidance.
Enhanced nav: True
```

Example 4 — Combined Conditions (and / or / not)

```
# and — BOTH must be True
is_night   = True
road       = 'Kafue Road'
```

```
danger_roads = ['Kafue Road', 'Great North Road']

if is_night and road in danger_roads:
    print('Night + dangerous road — ALERT sent!')

# or — at least ONE must be True
speed_kmh  = 108
fuel_pct   = 0.08

if speed_kmh > 100 or fuel_pct < 0.10:
    print('Urgent driver alert required!')

# not — reverses True/False
gps_connected = False
if not gps_connected:
    print('GPS signal lost — last known location used.')
```

```
Night + dangerous road — ALERT sent!
Urgent driver alert required!
GPS signal lost — last known location used.
```

Example 5 — Nested Conditions (Fleet Safety Gate)

```
vehicle_id = 'NZ-002'
driver_status = 'ON_DUTY'
speed_kmh     = 105.0
fuel_pct      = 0.45
gps_ok        = True

if driver_status == 'ON_DUTY':
    if gps_ok:
        if speed_kmh > 100:
            print(f'{vehicle_id}: SPEED ALERT — {speed_kmh} km/h')
        elif fuel_pct < 0.15:
            print(f'{vehicle_id}: LOW FUEL — {fuel_pct*100:.0f}%')
        else:
            print(f'{vehicle_id}: All systems normal.')
```

```
    else:
        print(f'{vehicle_id}: GPS offline!')
else:
    print(f'{vehicle_id}: Driver off duty — not tracking.')
```

```
NZ-002: SPEED ALERT — 105.0 km/h
```

✏ PRACTICE

1. Write if/elif/else: 'CRITICAL' if speed>120, 'WARNING' if speed>100, else 'OK'. Test 3 values.
2. Write night road check: RED if is_night AND road in danger_roads, AMBER if night only, else GREEN.
3. Write driver familiarity check: if driver not from Lusaka, enable enhanced_nav=True.
4. Use 'and': only alert if speed>100 AND is_night. Use 'or': alert if speed>100 OR fuel<10%.
5. Write nested ifs: check driver on_duty, then gps_ok, then speed > 100.

LESSON 06

Loops (for & while)

NAZON fleet scans all vehicles and all roads automatically every 30 seconds. Loops make that happen.

Example 1 — Basic for Loop: Scan All Vehicles

```
fleet = ['NZ-001','NZ-002','NZ-003','NZ-004']

for vehicle_id in fleet:
    print(f'Scanning {vehicle_id}...')
```

```
Scanning NZ-001...
Scanning NZ-002...
Scanning NZ-003...
Scanning NZ-004...
```

Example 2 — Loop with enumerate (Numbered Fleet Report)

```
fleet = ['NZ-001','NZ-002','NZ-003','NZ-004']

for i, vid in enumerate(fleet, 1):
    print(f'{i}. Checking {vid}')
```

```
1. Checking NZ-001
2. Checking NZ-002
3. Checking NZ-003
4. Checking NZ-004
```

Example 3 — Loop with Conditions (Real NAZON Road Scan)

```
roads = ['Great East Road','Cairo Road','Kafue Road','Mumbwa Road']
congestion_data = {
    'Great East Road': 7,
    'Cairo Road'     : 9,
    'Kafue Road'     : 4,
    'Mumbwa Road'    : 2,
}
alerts = {}

for road in roads:
    cong = congestion_data[road]
```

```
    if cong >= 8:
        alerts[road] = 'RED'
    elif cong >= 5:
        alerts[road] = 'AMBER'
    else:
        alerts[road] = 'GREEN'
    print(f'{road}: congestion={cong} → {alerts[road]}')
```

```
Great East Road: congestion=7 → AMBER
Cairo Road: congestion=9 → RED
Kafue Road: congestion=4 → GREEN
Mumbwa Road: congestion=2 → GREEN
```

Example 4 — while Loop (NAZON Live Tracking Loop)

```
import time

scan_count = 0
max_scans  = 3  # in real NAZON this runs forever (while True)

while scan_count < max_scans:
    scan_count += 1
    print(f'=== Scan #{scan_count} ===')
    print('  NZ-001: Great East Rd — 72 km/h — GREEN')
    print('  NZ-002: Cairo Road    — 38 km/h — RED (congestion)')
    print(f'  Next scan in 30 seconds...')
    print()

print('Max scans reached — NAZON Fleet stopped.')
```

```
=== Scan #1 ===
  NZ-001: Great East Rd — 72 km/h — GREEN
  NZ-002: Cairo Road    — 38 km/h — RED (congestion)
  Next scan in 30 seconds...

=== Scan #2 ===
  NZ-001: Great East Rd — 72 km/h — GREEN
```

```
  NZ-002: Cairo Road    — 38 km/h — RED (congestion)
  Next scan in 30 seconds...

Max scans reached — NAZON Fleet stopped.
```

Example 5 — break and continue

```
fleet = ['NZ-001','NZ-002','NZ-003','NZ-004']
no_gps = 'NZ-003'  # GPS offline

for vid in fleet:
    if vid == no_gps:
        print(f'Skipping {vid} — GPS offline')
        continue
    print(f'Analysed {vid} successfully')

print()
# Stop if critical vehicle breaks down
for vid in fleet:
    if vid == 'NZ-002':
        print(f'BREAKDOWN: {vid} — halting scan!')
        break
    print(f'Scanned {vid}')
```

```
Analysed NZ-001 successfully
Analysed NZ-002 successfully
Skipping NZ-003 — GPS offline
Analysed NZ-004 successfully

Scanned NZ-001
BREAKDOWN: NZ-002 — halting scan!
```

✏ PRACTICE

1. Loop through 4 vehicle IDs and print 'Pinging [ID]...' for each.
2. Create congestion dict for 4 roads. Loop and print RED/AMBER/GREEN for each.
3. Use enumerate to number vehicles: '1. NZ-001, 2. NZ-002...'
4. Use a while loop that runs 3 times, printing 'Scan #X — All vehicles checked'.

5. Loop through fleet but skip NZ-003 (GPS offline) using continue.

LESSON 07

Functions (Deep Dive)

Every NAZON Fleet module is a collection of functions. Master this and you can build anything.

Example 1 — Basic Function

```
def startup_check():
    print('NAZON Fleet Tracking System — Online')
    print('GPS modules: ACTIVE')
    print('Road intelligence: LOADED')

startup_check()
```

```
NAZON Fleet Tracking System — Online
GPS modules: ACTIVE
Road intelligence: LOADED
```

Example 2 — Function with Parameters

```
def check_vehicle(vehicle_id, road, speed_kmh):
    print(f'Vehicle : {vehicle_id}')
    print(f'Road    : {road}')
    print(f'Speed   : {speed_kmh} km/h')

check_vehicle('NZ-001', 'Great East Road', 65.5)
check_vehicle('NZ-002', 'Cairo Road', 32.0)
```

```
Vehicle : NZ-001
Road    : Great East Road
Speed   : 65.5 km/h
Vehicle : NZ-002
Road    : Cairo Road
Speed   : 32.0 km/h
```

Example 3 — Return Values

```
def get_danger_level(is_night, congestion, road, danger_roads):
    if is_night and road in danger_roads:
        return 'RED'
    elif congestion >= 8 or is_night:
        return 'AMBER'
```

```
    return 'GREEN'


def get_eta(distance_km, speed_kmh):
    if speed_kmh == 0:
        return None
    return round((distance_km / speed_kmh) * 60, 1)


danger = ['Kafue Road','Great North Road']
level = get_danger_level(True, 4, 'Kafue Road', danger)
print(f'Danger level: {level}')


eta = get_eta(45.0, 72.0)
print(f'ETA: {eta} minutes')
```

```
Danger level: RED
ETA: 37.5 minutes
```

Example 4 — The Speed Compliance Checker (Real NAZON Code)

```
# core/vehicle_monitor.py
def check_speed(vehicle_id, speed_kmh, road_type='highway'):
    '''
    Check if vehicle is within speed limit.
    road_type: highway=120, urban=80, residential=60
    '''
    limits = {'highway':120, 'urban':80, 'residential':60}
    limit  = limits.get(road_type, 100)
    over   = speed_kmh - limit

    if over > 20:
        return 'CRITICAL', f'Over by {over:.0f} km/h!'
    elif over > 0:
        return 'WARNING', f'Over by {over:.0f} km/h'
    return 'OK', 'Speed within limit'

status, msg = check_speed('NZ-003', 108, 'urban')
print(f'NZ-003: {status} — {msg}')
```

```
status, msg = check_speed('NZ-001', 115, 'highway')
print(f'NZ-001: {status} — {msg}')
```

```
NZ-003: CRITICAL — Over by 28 km/h!
NZ-001: OK — Speed within limit
```

Example 5 — Default Parameters + Multiple Returns

```
def analyse_vehicle(vehicle_id, speed_kmh, fuel_pct=1.0,
                    is_night=False, road='Unknown'):
    speed_ok  = speed_kmh <= 100
    fuel_ok   = fuel_pct  >= 0.15
    night_ok  = not is_night or speed_kmh <= 80
    safe      = speed_ok and fuel_ok and night_ok
    alert     = 'GREEN' if safe else 'AMBER'
    return safe, alert, speed_kmh, fuel_pct

safe, alert, spd, fuel = analyse_vehicle(
    'NZ-004', speed_kmh=95, fuel_pct=0.08, is_night=True)
print(f'Safe  : {safe}')
print(f'Alert : {alert}')
print(f'Speed : {spd} km/h')
print(f'Fuel  : {fuel*100:.0f}%')
```

```
Safe  : False
Alert : AMBER
Speed : 95 km/h
Fuel  : 8%
```

✏ PRACTICE

1. Write get_danger_level(is_night, congestion, road, danger_roads) returning RED/AMBER/GREEN.
2. Write get_eta(distance_km, speed_kmh) returning ETA in minutes as a float.
3. Write check_speed(vehicle_id, speed_kmh, road_type) returning status and message.
4. Write is_familiar_road(driver_city, current_roads_list) returning True/False.
5. Write a function returning 3 values: alert_level, advice, eta.

LESSON 08

Libraries & Imports

NAZON Fleet is powered by external libraries for GPS, maps, AI, HTTP requests and more.

Example 1 — Three Ways to Import

```
# Way 1: import whole library with alias

import pandas as pd

import numpy as np

import requests


# Way 2: import specific function only

from datetime import datetime

from os import getenv

from math import radians, sin, cos, sqrt, atan2


# Way 3: import everything (use carefully)

from math import *


# Using datetime:

now = datetime.now()

print(f'NAZON Fleet started at: {now.strftime("%Y-%m-%d %H:%M")}')
```

```
NAZON Fleet started at: 2026-05-12 21:45
```

Example 2 — datetime (Trip Logs, Night Detection)

```
from datetime import datetime


now = datetime.now()

print(now.strftime('%Y-%m-%d %H:%M:%S'))

print(now.strftime('%d/%m/%Y'))


# Is it night? (18:00 – 06:00)

hour = now.hour

is_night = hour >= 18 or hour < 6

print(f'Hour    : {hour}')

print(f'Is night: {is_night}')


# Format timestamp for SMS alert
```

```
alert_time = now.strftime('%d %b %Y at %H:%M')

print(f'Alert sent: {alert_time}')
```

```
2026-05-12 21:45:30

12/05/2026

Hour    : 21

Is night: True

Alert sent: 12 May 2026 at 21:45
```

Example 3 — math: Haversine Distance Between Two GPS Points

```
from math import radians, sin, cos, sqrt, atan2


def haversine_km(lat1, lon1, lat2, lon2):

    '''Calculate straight-line distance between two GPS points.'''

    R = 6371.0  # Earth radius in km

    dlat = radians(lat2 - lat1)

    dlon = radians(lon2 - lon1)

    a = sin(dlat/2)**2 + cos(radians(lat1)) * cos(radians(lat2)) * sin(dlon/2)**2

    c = 2 * atan2(sqrt(a), sqrt(1 - a))

    return round(R * c, 2)


# Lusaka CBD to Kenneth Kaunda Airport

dist = haversine_km(-15.4167, 28.2833, -15.3308, 28.4526)

print(f'Distance: {dist} km')
```

```
Distance: 21.47 km
```

Example 4 — requests: Call a Live Traffic API

```
# pip install requests

import requests


# Google Maps Directions API (real usage requires API key)

def get_traffic_duration(origin, destination, api_key):

    url = 'https://maps.googleapis.com/maps/api/directions/json'

    params = {

        'origin'      : origin,

        'destination' : destination,
```

```
        'departure_time': 'now',
        'key'          : api_key,
    }
    response = requests.get(url, params=params)
    data = response.json()
    leg = data['routes'][0]['legs'][0]
    return leg['duration_in_traffic']['text']

# Example call (requires real API key)
# eta = get_traffic_duration('Lusaka CBD', 'Airport', API_KEY)
# print(f'ETA in traffic: {eta}')
print('requests library ready — connect your Maps API key!')
```

```
requests library ready — connect your Maps API key!
```

Example 5 — pandas: Load and Analyse Incident Data

```
import pandas as pd

# Simulate incident data (in real NAZON from SQLite)
data = {
    'date'    : ['2026-05-01','2026-05-02','2026-05-03','2026-05-04'],
    'road'    : ['Kafue Rd','Cairo Rd','Great East Rd','Kafue Rd'],
    'type'    : ['pothole','congestion','accident','pothole'],
    'severity': [2, 4, 5, 3],
}
df = pd.DataFrame(data)
print(df)

print(f"\nKafue Rd incidents: {len(df[df['road']=='Kafue Rd'])}")
print(f"Avg severity: {df['severity'].mean():.1f}")
```

```
         date           road        type  severity
0  2026-05-01       Kafue Rd     pothole         2
1  2026-05-02       Cairo Rd  congestion         4
2  2026-05-03  Great East Rd    accident         5
3  2026-05-04       Kafue Rd     pothole         3
```

```
Kafue Rd incidents: 2

Avg severity: 3.5
```

■ Installing Libraries

Open your VS Code terminal (Ctrl+`) and run: pip install requests pandas numpy anthropic python-dotenv geopy folium schedule streamlit plotly

═ PRACTICE

1. Import datetime. Print today's date as DD/MM/YYYY and current hour.
2. Use the haversine_km function to measure Cairo Road CBD to UNZA distance.
3. Import pandas and create a DataFrame of 4 vehicles with speed and road columns.
4. Install requests. Print 'requests ready!' to confirm.
5. BONUS: Use datetime to detect if current hour is night (>=18 or <6). Print True/False.

INTERMEDIATE

Building NAZON Fleet's Core Modules You know enough Python to read real code. Now build the actual system.

LESSON 09

Classes & Object-Oriented Programming

Each NAZON Fleet module is a Class: VehicleTracker, RoadAnalyser, AlertSystem, AIAdvisor.

Example 1 — VehicleTracker Class

```
class VehicleTracker:

    def __init__(self, vehicle_id, driver, plate):

        self.vehicle_id = vehicle_id

        self.driver     = driver

        self.plate      = plate

        self.speed_kmh  = 0.0

        self.road       = None

        self.active     = False


    def start_trip(self, road):

        self.active = True

        self.road   = road

        print(f'{self.vehicle_id} ({self.driver}) started on {road}')


    def update_speed(self, speed_kmh):

        self.speed_kmh = speed_kmh

        if speed_kmh > 100:

            print(f'ALERT: {self.vehicle_id} — {speed_kmh} km/h!')


    def status(self):

        state = 'ACTIVE' if self.active else 'INACTIVE'

        print(f'{self.vehicle_id}: {state} | {self.speed_kmh} km/h | {self.road}')


v1 = VehicleTracker('NZ-001','Mwila Banda','ABZ 1234 ZM')

v1.status()

v1.start_trip('Great East Road')

v1.update_speed(107.5)

v1.status()
```

```
NZ-001: INACTIVE | 0.0 km/h | None

NZ-001 (Mwila Banda) started on Great East Road
```

```
ALERT: NZ-001 — 107.5 km/h!

NZ-001: ACTIVE | 107.5 km/h | Great East Road
```

Example 2 — RoadAnalyser Class

```
# core/road_analyser.py

class RoadAnalyser:

    DANGER_ROADS_NIGHT = [

        'Kafue Road','Great North Road',

        'Mumbwa Road','Lusaka-Livingstone',

        'Ndola-Kitwe Road',

    ]


    def __init__(self):

        self.congestion_data = {}


    def update_congestion(self, road, level):

        self.congestion_data[road] = level


    def get_danger(self, road, is_night, congestion):

        if is_night and road in self.DANGER_ROADS_NIGHT:

            return 'RED'

        elif congestion >= 8:

            return 'RED'

        elif congestion >= 5 or is_night:

            return 'AMBER'

        return 'GREEN'


    def get_advice(self, danger, road, is_night):

        if danger == 'RED' and is_night:

            return f'Avoid {road} at night. Use alternative route.'

        elif danger == 'RED':

            return f'Heavy congestion on {road}. Allow extra 30 mins.'

        elif danger == 'AMBER':

            return f'Caution on {road}. Reduce speed.'
```

```
        return f'{road} is clear. Safe travels.'


ra = RoadAnalyser()

d = ra.get_danger('Kafue Road', is_night=True, congestion=4)

a = ra.get_advice(d, 'Kafue Road', True)

print(d)

print(a)
```

```
RED

Avoid Kafue Road at night. Use alternative route.
```

Example 3 — AlertSystem Class

```
# core/alert_system.py

class AlertSystem:

    def __init__(self):

        self.alerts = []


    def send_alert(self, vehicle_id, level, message):

        alert = {

            'vehicle': vehicle_id,

            'level'  : level,

            'message': message,

        }

        self.alerts.append(alert)

        icon = {'RED':'■','AMBER':'■','GREEN':'■'}.get(level,'■')

        print(f'{icon} [{level}] {vehicle_id}: {message}')


    def get_red_alerts(self):

        return [a for a in self.alerts if a['level'] == 'RED']


    def summary(self):

        print(f'Total alerts : {len(self.alerts)}')

        print(f'RED alerts   : {len(self.get_red_alerts())}')


alerts = AlertSystem()

alerts.send_alert('NZ-001','RED','Night drive on Kafue Road — danger!')
```

```
alerts.send_alert('NZ-002','AMBER','Heavy congestion on Cairo Road')
alerts.send_alert('NZ-003','GREEN','Clear route — Mumbwa Road')
alerts.summary()
```

```
[RED] NZ-001: Night drive on Kafue Road — danger!
[AMBER] NZ-002: Heavy congestion on Cairo Road
[GREEN] NZ-003: Clear route — Mumbwa Road
Total alerts : 3
RED alerts   : 1
```

✏ PRACTICE

1. Create class VehicleTracker with vehicle_id, driver, road, speed. Add start_trip() and status().
2. Create class RoadAnalyser with get_danger() returning RED/AMBER/GREEN.
3. Create class AlertSystem with send_alert() and summary() methods.
4. Instantiate all 3 and call every method.
5. BONUS: Add block_vehicle() to VehicleTracker that sets active=False if fuel_pct<0.05.

LESSON 10

Error Handling (try / except)

NAZON Fleet never crashes — GPS dropouts, API failures, and bad data are all handled gracefully.

Example 1 — Basic try/except

```
# Without error handling — crashes the whole app
# float('bad gps data')  # ValueError

# With error handling
raw = 'bad gps data'
try:
    lat = float(raw)
except ValueError:
    print(f'Invalid GPS value: {raw!r}')
    lat = None

print(f'Latitude: {lat}')
print('App still running!')
```

```
Invalid GPS value: 'bad gps data'
Latitude: None
App still running!
```

Example 2 — Safe GPS Ping Parser

```
def parse_gps_ping(raw_data):
    '''Parse raw GPS data safely — never crash on bad data.'''
    try:
        lat = float(raw_data['lat'])
        lon = float(raw_data['lon'])
        spd = float(raw_data['speed'])
        return lat, lon, spd
    except (KeyError, ValueError, TypeError) as e:
        print(f'GPS parse error: {e}')
        return None, None, None

# Good ping
```

```
good = {'lat':'-15.4167','lon':'28.2833','speed':'65.5'}

print(parse_gps_ping(good))


# Corrupted ping

bad = {'lat':'LOST','lon':'28.2833'}

print(parse_gps_ping(bad))
```

```
(-15.4167, 28.2833, 65.5)

GPS parse error: could not convert string to float: 'LOST'

(None, None, None)
```

Example 3 — try / except / else / finally

```
def scan_vehicle(vehicle_id):

    valid_fleet = ['NZ-001','NZ-002','NZ-003','NZ-004']

    try:

        print(f'Scanning {vehicle_id}...')

        if vehicle_id not in valid_fleet:

            raise ValueError(f'{vehicle_id} not in NAZON Fleet')

        result = 'GREEN'

    except ValueError as e:

        print(f'Fleet error: {e}')

        result = None

    except Exception as e:

        print(f'Unexpected error: {e}')

        result = None

    else:

        print(f' Status: {result}')  # only if no exception

    finally:

        print(f' Scan of {vehicle_id} complete.')  # always runs

    return result


scan_vehicle('NZ-001')

print()

scan_vehicle('NZ-999')
```

```
Scanning NZ-001...
 Status: GREEN
 Scan of NZ-001 complete.

Scanning NZ-999...
Fleet error: NZ-999 not in NAZON Fleet
 Scan of NZ-999 complete.
```

✏ PRACTICE

1. Wrap float('bad gps') in try/except ValueError. Print friendly message.
2. Write parse_gps_ping() that raises ValueError on missing lat/lon keys.
3. Add else (success log) and finally (always log) to scan_vehicle().
4. Write safe_api_call() that catches requests.exceptions.ConnectionError.
5. Chain ValueError and generic Exception handlers in one try block.

LESSON 11

Working with Files & CSV

NAZON Fleet logs every GPS ping, incident and driver alert to files for review and analysis.

Example 1 — Write and Read a Trip Log

```
with open('nazon_trip.txt', 'w') as f:
    f.write('NAZON Fleet Trip Log\n')
    f.write('====================\n')
    f.write('NZ-001  Great East Road  65 km/h  GREEN\n')
    f.write('NZ-002  Cairo Road       32 km/h  RED\n')

with open('nazon_trip.txt', 'r') as f:
    content = f.read()
    print(content)
```

```
NAZON Fleet Trip Log
====================
NZ-001  Great East Road  65 km/h  GREEN
NZ-002  Cairo Road       32 km/h  RED
```

Example 2 — CSV Trip Journal

```
import csv
from datetime import datetime

def log_trip_event(vehicle_id, driver, road, speed, alert):
    with open('fleet_log.csv', 'a', newline='') as f:
        writer = csv.writer(f)
        writer.writerow([
            datetime.now().strftime('%Y-%m-%d %H:%M'),
            vehicle_id, driver, road,
            speed, alert
        ])
    print(f'Logged: {vehicle_id} on {road}')

log_trip_event('NZ-001','Mwila','Great East Rd',65,'GREEN')
log_trip_event('NZ-002','Chanda','Cairo Rd',32,'RED')
```

```
log_trip_event('NZ-003','Bwalya','Kafue Rd',88,'AMBER')


with open('fleet_log.csv','r') as f:
    for row in csv.reader(f):
        print(row)
```

```
['2026-05-12 21:45', 'NZ-001', 'Mwila', 'Great East Rd', '65', 'GREEN']
['2026-05-12 21:45', 'NZ-002', 'Chanda', 'Cairo Rd', '32', 'RED']
['2026-05-12 21:45', 'NZ-003', 'Bwalya', 'Kafue Rd', '88', 'AMBER']
```

✏ PRACTICE

1. Create fleet_alerts.txt. Write one line per vehicle with ID, road, and alert level.
2. Read the file back and print each line with its line number.
3. Write log_trip_event() that appends to fleet_log.csv with timestamp, id, road, speed, alert.
4. Log 3 events, read them all back, print each row.
5. BONUS: Read fleet_log.csv and count how many RED vs GREEN vs AMBER events.

LESSON 12

pandas & DataFrames

Fleet GPS logs come as pandas DataFrames. Learn to slice, filter and analyse them.

Example 1 — Fleet GPS Log as DataFrame

```
import pandas as pd


data = {

    'timestamp' : ['21:00','21:15','21:30','21:45','22:00'],

    'vehicle_id': ['NZ-001','NZ-002','NZ-001','NZ-003','NZ-002'],

    'road'      : ['Great East','Cairo Rd','Kafue Rd','Great North','Cairo Rd'],

    'speed_kmh' : [65.0, 28.0, 92.0, 78.0, 35.0],

    'alert'     : ['GREEN','RED','AMBER','GREEN','RED'],

}

df = pd.DataFrame(data)

print(df)
```

```
  timestamp vehicle_id        road  speed_kmh  alert

0     21:00     NZ-001  Great East       65.0  GREEN

1     21:15     NZ-002    Cairo Rd       28.0    RED

2     21:30     NZ-001    Kafue Rd       92.0  AMBER

3     21:45     NZ-003  Great North      78.0  GREEN

4     22:00     NZ-002    Cairo Rd       35.0    RED
```

Example 2 — Filtering & Analysis

```
# All RED alert events

red_events = df[df['alert'] == 'RED']

print(red_events)


# All speeds above 85 km/h

speeding = df[df['speed_kmh'] > 85]

print(f'Speeding events: {len(speeding)}')


# Average speed per vehicle

print(df.groupby('vehicle_id')['speed_kmh'].mean().round(1))
```

```
  timestamp vehicle_id      road  speed_kmh alert
```

```
1     21:15     NZ-002  Cairo Rd       28.0   RED

4     22:00     NZ-002  Cairo Rd       35.0   RED


Speeding events: 1


vehicle_id

NZ-001    78.5

NZ-002    31.5

NZ-003    78.0
```

✏ PRACTICE

1. Create a DataFrame with 5 GPS log rows (timestamp, vehicle_id, road, speed_kmh, alert).
2. Filter for all RED alerts. Print them.
3. Find the maximum speed recorded and which vehicle it belongs to.
4. Group by vehicle_id and print average speed for each.
5. Add a 'speeding' column: True if speed_kmh > 100.

LESSON 13

.env Files & Config Management

NAZON Fleet keeps API keys and server credentials secure using .env files.

Never hardcode your Google Maps API key, SMS gateway password or database URL in code. Use .env files — they stay off GitHub and off every device that should not have them.

Step 1 — Create the .env File

```
# Create a file called: .env  (project root folder)

# This is NOT Python — just key=value lines:


GOOGLE_MAPS_KEY=AIzaSy_your_real_key_here

CLAUDE_API_KEY=sk-ant-your_key_here

SMS_API_KEY=your_sms_gateway_key

FLEET_DB=nazon_fleet.db

SPEED_LIMIT_HIGHWAY=120

SPEED_LIMIT_URBAN=80

NIGHT_HOUR_START=18

NIGHT_HOUR_END=6
```

Step 2 — config.py (Central Settings File)

```
# config.py

import os

from dotenv import load_dotenv


load_dotenv()


# Secrets (from .env)

GOOGLE_MAPS_KEY   = os.getenv('GOOGLE_MAPS_KEY')

CLAUDE_API_KEY    = os.getenv('CLAUDE_API_KEY')

SMS_API_KEY       = os.getenv('SMS_API_KEY')

FLEET_DB          = os.getenv('FLEET_DB', 'nazon_fleet.db')


# Speed limits

SPEED_LIMIT_HWY   = int(os.getenv('SPEED_LIMIT_HIGHWAY', '120'))

SPEED_LIMIT_URBAN = int(os.getenv('SPEED_LIMIT_URBAN', '80'))


# Night detection
```

```
NIGHT_START = int(os.getenv('NIGHT_HOUR_START', '18'))
NIGHT_END   = int(os.getenv('NIGHT_HOUR_END',   '6'))

# Fleet config (not secret — hardcoded is fine)
FLEET_IDS = ['NZ-001','NZ-002','NZ-003','NZ-004','NZ-005']
SCAN_INTERVAL_SEC = 30
DANGER_ROADS_NIGHT = [
    'Kafue Road','Great North Road',
    'Mumbwa Road','Lusaka-Livingstone','Ndola-Kitwe Road',
]

print(f'Google Maps key set: {bool(GOOGLE_MAPS_KEY)}')
print(f'Fleet size         : {len(FLEET_IDS)}')
print(f'Night hours        : {NIGHT_START}:00 – {NIGHT_END}:00')
```

```
Google Maps key set: True
Fleet size         : 5
Night hours        : 18:00 – 6:00
```

■ Never upload .env to GitHub!

Anyone with your Google Maps key can rack up thousands of dollars in API charges. Anyone with your SMS key can send messages billed to you. Add .env to .gitignore immediately.

⌨ PRACTICE

1. Create a .env with GOOGLE_MAPS_KEY, CLAUDE_API_KEY, FLEET_DB.
2. Create config.py that loads and prints them (show key as bool, not value).
3. Add FLEET_IDS, SCAN_INTERVAL_SEC, DANGER_ROADS_NIGHT to config.py.
4. Create a second file that imports from config and loops through FLEET_IDS.
5. Create .gitignore and add .env, *.db, __pycache__/ to it.

LESSON 14

APIs & the Claude AI Road Intelligence

NAZON Fleet sends live road data to Claude AI and receives driver safety advice in real-time.

Example 1 — The NAZON Fleet AI Flow

```
# The flow:

# 1. GPS ping arrives — vehicle_id, road, speed, lat/lon

# 2. RoadAnalyser calculates congestion and danger level

# 3. NAZON builds a prompt describing the situation

# 4. Claude AI receives the prompt

# 5. Claude replies: ROAD_STATUS, SAFETY_LEVEL, DRIVER_ADVICE

# 6. NAZON sends the advice to the driver via SMS / dashboard
```

Example 2 — Simple Claude API Call

```
import anthropic

import os

from dotenv import load_dotenv

load_dotenv()


client = anthropic.Anthropic(api_key=os.getenv("CLAUDE_API_KEY"))


response = client.messages.create(

    model      = 'claude-sonnet-4-20250514',

    max_tokens = 100,

    messages   = [

        {'role':'user', 'content': 'Say hello from NAZON Fleet!'}

    ]

)

print(response.content[0].text)
```

```
Hello from NAZON Fleet! Ready to keep your drivers safe on Zambian roads.
```

Example 3 — The AI Road Advisor Class

```
import anthropic

from config import CLAUDE_API_KEY


class AIRoadAdvisor:
```

```
    def __init__(self):
        self.client = anthropic.Anthropic(api_key=CLAUDE_API_KEY)

    def advise(self, vehicle_id, road, speed_kmh,
               congestion, is_night, driver_familiar, destination):
        prompt = f'''
        You are NAZON Fleet AI — a road safety assistant for Zambia.
        A vehicle needs a safety advisory right now.

        Vehicle    : {vehicle_id}
        Road       : {road}
        Speed      : {speed_kmh} km/h
        Congestion : {congestion}/10
        Night drive: {is_night}
        Driver familiar with road: {driver_familiar}
        Destination: {destination}

        Reply ONLY in this format:
        ROAD_STATUS: [CLEAR/CONGESTED/DANGEROUS]
        SAFETY_LEVEL: [GREEN/AMBER/RED]
        ADVICE: [one sentence for the driver]
        '''
        r = self.client.messages.create(
            model='claude-sonnet-4-20250514',
            max_tokens=80,
            messages=[{'role':'user','content':prompt}]
        )
        return r.content[0].text.strip()
```

Example 4 — Using the Advisor

```
advisor = AIRoadAdvisor()
result = advisor.advise(
    vehicle_id      = 'NZ-003',
    road            = 'Kafue Road',
```

```
    speed_kmh       = 85.0,
    congestion      = 3,
    is_night        = True,
    driver_familiar = False,
    destination     = 'Kafue Town'
)
print(result)
```

```
ROAD_STATUS: DANGEROUS
SAFETY_LEVEL: RED
ADVICE: Kafue Road is poorly lit at night with potholes — reduce speed to 60 km/h and use full be
am headlights.
```

✏ PRACTICE

1. Get your free Claude API key from console.anthropic.com.
2. Add CLAUDE_API_KEY to your .env file.
3. Write a script that sends 'Analyse NZ-001 on Cairo Road: congestion 9/10' to Claude.
4. Print Claude's full response.
5. BONUS: Build the full AIRoadAdvisor class and test it with 3 different scenarios.

P

PRO

Building NAZON Fleet End-to-End This is where everything comes together. Real production-grade fleet code.

LESSON 15

Advanced OOP & Module Design

How all NAZON Fleet modules connect, inherit and communicate with each other.

Example 1 — BaseModule Inheritance

```
class BaseModule:

    def __init__(self, name):

        self.name   = name

        self.active = False


    def start(self):

        self.active = True

        print(f'{self.name} started')


    def stop(self):

        self.active = False

        print(f'{self.name} stopped')


    def status(self):

        state = 'ACTIVE' if self.active else 'INACTIVE'

        print(f'{self.name}: {state}')


class GPSReceiver(BaseModule):

    def __init__(self):

        super().__init__('GPS Receiver')

        self.vehicle_count = 0


    def register_vehicle(self, vid):

        self.vehicle_count += 1

        print(f'Registered {vid} ({self.vehicle_count} total)')


gps = GPSReceiver()

gps.status()

gps.start()

gps.register_vehicle('NZ-001')

gps.register_vehicle('NZ-002')
```

```
gps.status()
```

```
GPS Receiver: INACTIVE

GPS Receiver started

Registered NZ-001 (1 total)

Registered NZ-002 (2 total)

GPS Receiver: ACTIVE
```

Example 2 — main.py Wires All Modules Together

```
# main.py — NAZON Fleet conductor

from config import FLEET_IDS, SCAN_INTERVAL_SEC, DANGER_ROADS_NIGHT

# from core.gps_receiver    import GPSReceiver

# from core.road_analyser   import RoadAnalyser

# from core.alert_system    import AlertSystem

# from ai.road_advisor      import AIRoadAdvisor

# from journal.trip_logger  import TripLogger


def run_nazon_fleet():

    print('=' * 45)

    print('  NAZON FLEET TRACKING SYSTEM — v1.0')

    print('=' * 45)


    for vid in FLEET_IDS:

        print(f'Pinging {vid}...')

        # gps_data = gps.get_ping(vid)

        # road_info = analyser.get_status(gps_data)

        # advice = ai_advisor.advise(vid, road_info)

        # alert_system.send_alert(vid, advice)

        # logger.log(vid, gps_data, advice)


    print('\nNAZON Fleet scan complete.')


run_nazon_fleet()
```

```
=============================================

  NAZON FLEET TRACKING SYSTEM — v1.0
```

```
==========================================

Pinging NZ-001...

Pinging NZ-002...

Pinging NZ-003...

NAZON Fleet scan complete.
```

LESSON 16

Decorators & Live Alert Logic

NAZON Fleet uses decorators to add logging and retry logic without repeating code.

Example 1 — Logging Decorator

```
def log_it(func):
    def wrapper(*args, **kwargs):
        print(f'>>> {func.__name__} called')
        result = func(*args, **kwargs)
        print(f'<<< {func.__name__} done')
        return result
    return wrapper

@log_it
def scan_vehicle(vid):
    print(f'  Scanning {vid}...')
    return 'GREEN'

alert = scan_vehicle('NZ-001')
print(f'Alert: {alert}')
```

```
>>> scan_vehicle called
  Scanning NZ-001...
<<< scan_vehicle done
Alert: GREEN
```

Example 2 — Retry Decorator (GPS Reconnect)

```
import time

def retry(attempts=3, delay=1):
    def decorator(func):
        def wrapper(*args, **kwargs):
            for i in range(attempts):
                try:
                    return func(*args, **kwargs)
                except Exception as e:
```

```
                    print(f'Attempt {i+1} failed: {e}')
                    if i < attempts - 1:
                        time.sleep(delay)
            print(f'{func.__name__} failed after {attempts} attempts')
            return None
        return wrapper
    return decorator

attempt_num = 0

@retry(attempts=3, delay=0)
def connect_gps():
    global attempt_num
    attempt_num += 1
    if attempt_num < 3:
        raise ConnectionError('GPS antenna not responding')
    print(f'GPS connected on attempt {attempt_num}')
    return True

connect_gps()
```

```
Attempt 1 failed: GPS antenna not responding
Attempt 2 failed: GPS antenna not responding
GPS connected on attempt 3
```

LESSON 17

Threading & Real-Time Tracking

NAZON Fleet tracks all vehicles and runs the dashboard simultaneously using threads.

Example 1 — schedule Library (30-Second GPS Scans)

```
# pip install schedule

import schedule, time


scan_count = 0


def scan_all_fleet():

    global scan_count

    scan_count += 1

    print(f'Scan #{scan_count}: NZ-001=GREEN NZ-002=RED NZ-003=AMBER')


# Real NAZON: schedule.every(30).seconds.do(scan_all_fleet)

schedule.every(2).seconds.do(scan_all_fleet)


scan_all_fleet()

for _ in range(4):

    schedule.run_pending()

    time.sleep(1)
```

```
Scan #1: NZ-001=GREEN NZ-002=RED NZ-003=AMBER

Scan #2: NZ-001=GREEN NZ-002=RED NZ-003=AMBER

Scan #3: NZ-001=GREEN NZ-002=RED NZ-003=AMBER
```

Example 2 — Background Thread

```
import threading, time


def background_gps_tracker():

    for i in range(3):

        print(f'[GPS] Ping #{i+1} — all vehicles located')

        time.sleep(1)

    print('[GPS] Done.')


thread = threading.Thread(target=background_gps_tracker, daemon=True)

thread.start()
```

```
for i in range(3):
    print(f'[DASHBOARD] Refreshing display #{i+1}')
    time.sleep(1)

thread.join()
print('Both finished!')
```

```
[DASHBOARD] Refreshing display #1
[GPS] Ping #1 — all vehicles located
[DASHBOARD] Refreshing display #2
[GPS] Ping #2 — all vehicles located
[DASHBOARD] Refreshing display #3
[GPS] Ping #3 — all vehicles located
[GPS] Done.
Both finished!
```

LESSON 18

SQLite Fleet Journey Log

NAZON Fleet logs every GPS ping, alert and incident to SQLite for analytics and review.

Example 1 — Create Database

```
import sqlite3


def init_fleet_db():

    conn = sqlite3.connect('nazon_fleet.db')

    c = conn.cursor()

    c.execute('''

        CREATE TABLE IF NOT EXISTS trips (

            id          INTEGER PRIMARY KEY AUTOINCREMENT,

            timestamp   TEXT NOT NULL,

            vehicle_id  TEXT NOT NULL,

            driver      TEXT,

            road        TEXT,

            speed_kmh   REAL,

            alert_level TEXT DEFAULT "GREEN",

            ai_advice   TEXT,

            is_night    INTEGER DEFAULT 0

        )

    ''')

    conn.commit()

    conn.close()

    print('Fleet database ready!')


init_fleet_db()
```

```
Fleet database ready!
```

Example 2 — Log and Query Trips

```
from datetime import datetime


def log_trip(vehicle_id, driver, road, speed, alert, advice, night):

    conn = sqlite3.connect('nazon_fleet.db')
```

```
    conn.cursor().execute(
        'INSERT INTO trips (timestamp,vehicle_id,driver,road,speed_kmh,alert_level,ai_advice,is_night)'
        ' VALUES (?,?,?,?,?,?,?,?)',
        (datetime.now().isoformat(), vehicle_id, driver,
         road, speed, alert, advice, int(night))
    )
    conn.commit()
    conn.close()

log_trip('NZ-001','Mwila','Great East Rd',65,'GREEN','Safe travels.',False)
log_trip('NZ-002','Chanda','Kafue Road',88,'RED','Dangerous at night.',True)

conn = sqlite3.connect('nazon_fleet.db')
rows = conn.execute('SELECT vehicle_id,road,alert_level FROM trips').fetchall()
conn.close()
for r in rows:
    print(r)
```

```
('NZ-001', 'Great East Rd', 'GREEN')
('NZ-002', 'Kafue Road', 'RED')
```

LESSON 19

Maps & Geospatial Deep Dive

NAZON Fleet visualises routes, plots danger zones and calculates real distances on Zambian roads.

Example 1 — Haversine: GPS Distance Calculator

```
from math import radians, sin, cos, sqrt, atan2


def haversine_km(lat1, lon1, lat2, lon2):

    R = 6371.0

    dlat = radians(lat2 - lat1)

    dlon = radians(lon2 - lon1)

    a = sin(dlat/2)**2 + cos(radians(lat1))*cos(radians(lat2))*sin(dlon/2)**2

    return round(R * 2 * atan2(sqrt(a), sqrt(1-a)), 2)


# Key Zambian distances

routes = {

    'Lusaka → Chipata'    : [(-15.4167,28.2833),(-13.6436,32.6444)],

    'Lusaka → Livingstone': [(-15.4167,28.2833),(-17.8418,25.8545)],

    'Lusaka → Ndola'      : [(-15.4167,28.2833),(-12.9587,28.6366)],

    'Lusaka → Solwezi'    : [(-15.4167,28.2833),(-12.1730,26.3927)],

}


for route, (p1, p2) in routes.items():

    d = haversine_km(*p1, *p2)

    print(f'{route}: {d} km')
```

```
Lusaka → Chipata    : 581.23 km

Lusaka → Livingstone: 469.51 km

Lusaka → Ndola      : 319.74 km

Lusaka → Solwezi    : 506.88 km
```

Example 2 — folium: Plot Fleet on a Live Map

```
# pip install folium

import folium


# Create map centred on Lusaka

m = folium.Map(location=[-15.4167, 28.2833], zoom_start=12)
```

```
# Vehicle positions
vehicles = [
    {'id':'NZ-001','lat':-15.4167,'lon':28.2833,'alert':'GREEN'},
    {'id':'NZ-002','lat':-15.3955,'lon':28.3050,'alert':'RED'},
    {'id':'NZ-003','lat':-15.4400,'lon':28.2600,'alert':'AMBER'},
]

colours = {'GREEN':'green','AMBER':'orange','RED':'red'}
for v in vehicles:
    colour = colours[v['alert']]
    folium.CircleMarker(
        location=[v['lat'], v['lon']],
        radius=10,
        color=colour,
        fill=True,
        popup=f"{v['id']} — {v['alert']}",
    ).add_to(m)

m.save('nazon_fleet_map.html')
print('Map saved as nazon_fleet_map.html')
```

```
Map saved as nazon_fleet_map.html
```

■ *Open nazon_fleet_map.html in any browser to see your live fleet map. In the full NAZON Fleet dashboard this updates every 30 seconds via Streamlit.*

✏ PRACTICE

1. Use haversine_km to calculate distance: Lusaka (-15.4167, 28.2833) → UNZA (-15.4177, 28.3122).
2. Create a folium map centred on Lusaka. Add 3 vehicle markers with different alert colours.
3. Save the map as fleet_map.html and open it in a browser.
4. Add a danger zone circle (folium.Circle, radius=500m, red) on Kafue Road.
5. BONUS: Add a route polyline from Lusaka CBD to the airport using folium.PolyLine.

LESSON 20

NAZON Fleet v1 — Full Working System

All 20 lessons combined into one complete, runnable fleet safety scanner.

This is NAZON Fleet v1. Every concept from all 20 lessons is present: variables, functions, classes, loops, error handling, pandas, Claude AI, SQLite and maps.

Part 1 — Imports, Config & Database Setup

```
# nazon_fleet_v1.py — Full NAZON Fleet Scanner

import sqlite3, os, time

import pandas as pd

from datetime import datetime

from math import radians, sin, cos, sqrt, atan2

from dotenv import load_dotenv

import anthropic


load_dotenv()

API_KEY = os.getenv("CLAUDE_API_KEY")


FLEET = [

    {'id':'NZ-001','driver':'Mwila Banda',  'road':'Great East Road','speed':65,'night':False},

    {'id':'NZ-002','driver':'Chanda Phiri', 'road':'Kafue Road',     'speed':88,'night':True},

    {'id':'NZ-003','driver':'Bwalya Mutale','road':'Cairo Road',     'speed':30,'night':False},

]


DANGER_ROADS = ["Kafue Road","Great North Road","Mumbwa Road"]


def init_db():

    conn = sqlite3.connect("nazon_fleet.db")

    conn.cursor().execute('''

        CREATE TABLE IF NOT EXISTS trips (

            id INTEGER PRIMARY KEY AUTOINCREMENT,

            timestamp TEXT, vehicle_id TEXT, driver TEXT,

            road TEXT, speed_kmh REAL, alert TEXT, advice TEXT)

    ''')

    conn.commit(); conn.close()
```

Part 2 — Core Functions

```
def get_danger(road, speed_kmh, is_night):
    if is_night and road in DANGER_ROADS:
        return "RED"
    elif speed_kmh > 100 or (is_night and speed_kmh > 80):
        return "RED"
    elif speed_kmh > 80 or is_night:
        return "AMBER"
    return "GREEN"

def get_ai_advice(vehicle, danger):
    try:
        client = anthropic.Anthropic(api_key=API_KEY)
        prompt = f'''
        NAZON Fleet AI — Zambian road safety.
        Vehicle: {vehicle["id"]} on {vehicle["road"]}
        Speed: {vehicle["speed"]} km/h | Night: {vehicle["night"]}
        Danger: {danger}
        Reply: ADVICE: [one sentence for the driver]
        '''
        r = client.messages.create(
            model="claude-sonnet-4-20250514",
            max_tokens=60,
            messages=[{'role':'user','content':prompt}]
        )
        return r.content[0].text.strip()
    except Exception as e:
        return f"Stay alert on {vehicle["road"]}."

def log_trip(v, alert, advice):
    conn = sqlite3.connect("nazon_fleet.db")
    conn.cursor().execute(
        "INSERT INTO trips (timestamp,vehicle_id,driver,road,speed_kmh,alert,advice)"
```

```
        " VALUES (?,?,?,?,?,?,?)",
        (datetime.now().isoformat(),v["id"],v["driver"],
         v["road"],v["speed"],alert,advice)
    )
    conn.commit(); conn.close()
```

Part 3 — Main Scanner

```
def run():
    print('=' * 46)
    print('  NAZON FLEET TRACKING SYSTEM  v1.0')
    print('=' * 46)
    init_db()
    print()

    for v in FLEET:
        try:
            print(f"Scanning {v['id']} ({v['driver']})...")
            danger = get_danger(v['road'], v['speed'], v['night'])
            advice = get_ai_advice(v, danger)
            log_trip(v, danger, advice)
            print(f"  Road   : {v['road']}")
            print(f"  Speed  : {v['speed']} km/h")
            print(f"  Night  : {v['night']}")
            print(f"  Alert  : {danger}")
            print(f"  Advice : {advice}")
            print()
        except Exception as e:
            print(f"  Error on {v['id']}: {e}")

    print('NAZON Fleet scan complete!')

if __name__ == '__main__':
    run()
```

```
==============================================
```

```
  NAZON FLEET TRACKING SYSTEM  v1.0
==========================================

Scanning NZ-001 (Mwila Banda)...
  Road   : Great East Road
  Speed  : 65 km/h
  Night  : False
  Alert  : GREEN
  Advice : Great East Road is clear — maintain speed and safe journey.

Scanning NZ-002 (Chanda Phiri)...
  Road   : Kafue Road
  Speed  : 88 km/h
  Night  : True
  Alert  : RED
  Advice : Kafue Road is dangerous at night — reduce to 60 km/h, full headlights on.

NAZON Fleet scan complete!
```

■ *NAZON Fleet v1 is complete. Phase 2 adds a Streamlit live dashboard. Phase 3 adds SMS driver alerts via AfricasTalking API. Phase 4 adds the folium live map with auto-refresh.*

QUICK REFERENCE — NAZON FLEET CHEATSHEET

Concept	Syntax / Example	Where in NAZON Fleet
Variable	`speed_kmh = 65.5`	Every module
f-string	`f'Alert: {road}'`	Alerts, AI prompts
List	`['NZ-001','NZ-002',...]`	Fleet IDs, danger roads
Dict	`{'id':'NZ-001','speed':65}`	Vehicles, GPS pings, roads
if/elif/else	`if is_night and road in danger:`	road_analyser.py
for loop	`for v in FLEET:`	Scanning all vehicles
while loop	`while tracking:`	Main scan loop
function	`def get_danger(road,...):`	All modules
class	`class RoadAnalyser:`	Each module file
try/except	`try: ... except:`	GPS calls, API calls
with open	`with open('log.csv','a'):`	Trip journal
pandas	`df[df['alert']=='RED']`	Fleet analytics
haversine	`haversine_km(lat1,lon1,...)`	Distance calculation
folium	`folium.Map(location=[lat,lon])`	Live map dashboard
Claude API	`client.messages.create()`	ai/road_advisor.py
.env / getenv	`os.getenv('GOOGLE_MAPS_KEY')`	config.py
SQLite	`cursor.execute('INSERT...')`	journal/trip_logger.py
schedule	`schedule.every(30).seconds.do()`	main.py loop
threading	`Thread(target=gps_scan)`	Parallel GPS + dashboard



You now have everything you need to build a live fleet safety system that advises every driver on every dangerous road in Lusaka, Zambia and Africa. Practice every example. Break the code. Fix it. That is how real developers learn. When NAZON Fleet v1 runs and your first RED night-road alert reaches a driver on Kafue Road, you will know it was worth it. "The future depends on what you do today."